

OPTIONS FROM ZERO • PART 3

Options from Zero Part 3

Beyond Black-Scholes: the smile,
local vol, Heston and SABR



Rates with Bip

OPTIONS FROM ZERO • PART 3 OF 3

Options from Zero

Part 3: Beyond Black-Scholes, from the smile to local vol, Heston and SABR, with Python and QuantLib

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Options from Zero. Part 3: Beyond Black-Scholes, from the smile to local vol, Heston and SABR, with Python and QuantLib.

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The share in Chapters 1-3, its option prices, its volatility and the interest rate are illustrative, made up for teaching, and every simulated path is simulated. The AUD/USD volatility quotes, the AUD/USD, SOFR, AONIA and BBSW quotes (the illustrative market of the Series 2 book), the caplet smile in Chapter 9 and the options book in Chapter 10 are illustrative too. They are not market data. The AUD/USD exchange rates in Chapter 4 are the noon buying rates in New York from the Federal Reserve Board's H.10 release. Source: Board of Governors of the Federal Reserve System.

AUD/USD option conventions are checked against Reiswich and Wystup (2009), the April 2025 episode against the Reserve Bank of Australia's own account, and SVI and its arbitrage conditions against Gatheral and Jacquier (2014); sources are listed at the end of each chapter. Options are risky: a seller can lose far more than the premium received.

Every number in the text, tables and figures is computed by the companion QuantLib Python notebooks (QuantLib 1.43), one per chapter, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This book follows the ten video episodes of Part 3 of *Options from Zero*, one chapter per episode, and goes a little deeper in each. It picks up where Part 2 ended: you can derive Black-Scholes from a tree and from hedging, read the Greeks, and price FX and interest-rate options. Part 3 does two things.

- **Chapters 1–3** finish the story of the formula. An expectation is an area under a density; changing the odds (Girsanov) turns real-world paths into risk-neutral ones; and the tree, the PDE and the expectation turn out to be one theory (Feynman-Kac).
- **Chapters 4–9** break Black-Scholes and rebuild it. Real AUD/USD returns have fat tails, so the market charges a different volatility at every strike: the smile. We build an arbitrage-free surface with SVI, read the market's risk-neutral density from it, and fit the three model families desks use: local volatility, stochastic volatility (Heston) and SABR.
- **Chapter 10** runs a small AUD/USD options book: its risks, its hedges, and a week of P&L explained two ways.

Each chapter ends with the chapter's one idea to remember, which the videos call *Bip's rule*, and a question to try before the next chapter; the answer opens the next chapter. The appendix has the final checkpoint with worked answers, and a glossary.

Numbers you can reproduce

Every number in the text, tables and figures is computed by a companion Python notebook, one per chapter; nothing is typed in by hand. Simulations use fixed random seeds. Where the book works something by hand, the notebook checks it against QuantLib, and several of those checks caught real mistakes while this book was being made.

The notebooks are free to download from github.com/RatesWithBip/curriculum, in the `options-from-zero` folder, and each has an *Open in Colab* button. They run as they are in Google Colab: each installs QuantLib and writes its own data files.

About the data

The share of Chapters 1–3 is Parts 1 and 2's: \$100, no dividends, 20% volatility, a 3.60% interest rate. From Chapter 4 the examples are on AUD/USD, using the illustrative market of the *SOFR, FX & Cross-Currency Curves* book and an illustrative set of AUD/USD volatility quotes. Real option quotes are licensed data, and the ideas don't depend on the exact levels. The one exception is Chapter 4, which measures the fat tails of the Australian dollar from the Federal Reserve Board's published exchange rates.

Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

Options are risky. This book is for education only and is not financial advice.

An Expectation Is an Integral

A maths pit stop. A density's height isn't a probability; the area under it is. An option's price is the discounted area under payoff $\times$ density.

The density

Under the risk-neutral odds, our share's price in a year has the lognormal density of Part 1,

$$f(x) = \frac{1}{x \sigma \sqrt{T} \sqrt{2\pi}} \exp\left(-\frac{(\ln x - m)^2}{2\sigma^2 T}\right), \quad m = \ln S + (r - \sigma^2/2) T.$$

The height of the curve is not a probability; the **area** under it between two prices is the chance of ending up between them. So the whole area is 1 (the notebook gets 1.000000), and the curve's balance point, its mean, is the forward, \$103.67.

An average is an area

The expected value of anything that depends on the price is the area under that thing times the density:

$$\mathbb{E}[g(S_T)] = \int g(x) f(x) dx.$$

Picture thin vertical strips: each is a price, times its density, times the strip's width. For the call's payoff, discounted:

Payoff $\times$ density added up strip by strip from the strike to \$250, then discounted.

STRIPS	WIDTH (\$)	DISCOUNTED AREA
5	29.00	\$8.100
20	7.25	\$7.429
100	1.45	\$7.391
1,000	0.14	\$7.389

With thinner strips the sum settles on the exact area: the **integral**, written with a long S for *sum*.

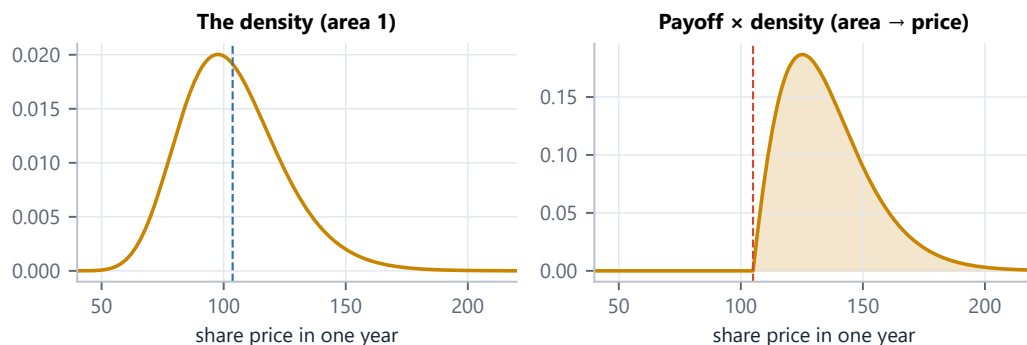


Figure 1.1. Left: the density, balancing at the forward. Right: payoff $\times$ density, whose area, discounted, is the price.

Payoff times density

$$C = e^{-rT} \int_K^{\infty} (x - K) f(x) dx = 7.3896$$

the Black-Scholes price to a few billionths of a dollar. The area of the density alone above the strike is 43.5%: that's $N(d_2)$.

Two ways to find an area

Part 2's Monte Carlo averaged a million simulated payoffs: $\$7.406 \pm 0.013$. That is the same area, estimated by sampling. Integrate when you know the density; simulate when you can only draw from it. For the rest of this book, the density is the bridge: change the density, and you change every price.

Bip's rule. Price = discounted area under payoff $\times$ density.

Try this. What does a density integrate to?

Companion notebook: `s3e22_expectation_integral.ipynb`. Every number in this chapter is computed by it.

Changing the Odds: Martingales and Girsanov

Simulate in the real world and still get the right price, by weighting each path. The weights turn the real drift into the risk-free rate and leave volatility alone.

Last chapter's question

One: all the probability there is.

The wrong way

Simulate 1,000,000 paths of our share in the real world, where it is expected to grow at 8% a year, average the call's payoffs and discount at the risk-free 3.60%: \$9.89, against Black-Scholes' \$7.39. The payoff is risky, far riskier than the share; discounting a risky average at the riskless rate overprices it. Real-world odds and a risk-free discount rate don't mix.

Martingales

A **martingale** is a process whose expected future value is its value today: a fair game, with no drift. In the real world our share, discounted at the risk-free rate, isn't one: its expected value in a year is \$104.48 in today's money. Under the risk-neutral odds it is. That is the precise meaning of *risk-neutral*: odds under which every price, discounted at the risk-free rate, is a martingale.

Reweighting the paths

Girsanov (1960) showed that changing the odds is the same as weighting each path by

$$L = \exp \left(-\lambda W_T - \frac{1}{2} \lambda^2 T \right), \quad \lambda = \frac{\mu - r}{\sigma} = 0.22,$$

where W_T is the path's Brownian motion and λ the **market price of risk**, the extra expected return per unit of volatility. The weights average to 1 (1.0002 over a million paths), turn the drift into r , and leave the volatility unchanged.

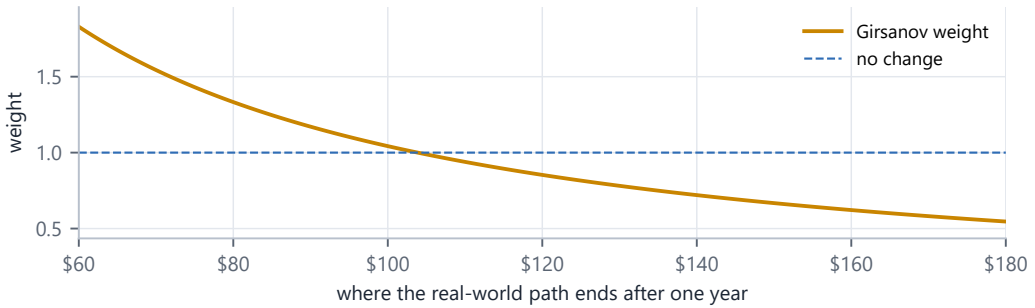


Figure 2.1. A path's weight against where it ends. High endings are shrunk, low ones swollen.

A path ending at \$150 counts for 0.67 of a path; one ending at \$80, 1.33.

Same paths, right price

One million real-world paths, used two ways.

	PRICE	CHANCE OF EXERCISE
Real-world odds, discounted at r	\$9.885	52.3%
Same paths, reweighted	\$7.381	43.5%
Black-Scholes	\$7.390	43.5%

The reweighted average is the Black-Scholes price within its error bar (± 0.010), and the reweighted chance of exercise is $N(d_2)$.

Which busts a stubborn myth: risk-neutral pricing does not assume investors ignore risk. It is a different set of odds, chosen so that a single discount rate works for everything.

Bip's rule. Change the odds, not the paths: drift becomes r , volatility stays.

Try this. If the share's expected return equalled the risk-free rate, what would every path's weight be?

Sources

Companion notebook: `s3e23_change_of_measure.ipynb`. Every number in this chapter is computed by it.

1. Math-Net.Ru, *On transforming a certain class of stochastic processes by absolutely continuous substitution of measures* (I. V. Girsanov), *Teoriya Veroyatnostei i ee Primeneniya* 5(3), 314-330 (1960); English translation in *Theory of Probability and Its Applications* 5(3) (1960). <https://www.mathnet.ru/eng/tvp4837>

Three Roads, One Price

Replicating on a tree, hedging with a PDE and averaging under risk-neutral odds are one theory, tied together by Feynman-Kac.

Last chapter's question

Exactly one. With $\mu = r$ the market price of risk is zero, and the two sets of odds coincide.

The same trade, three ways

The one-year \$105 call, priced along each of the series' three roads.

		PRICE	VS FORMULA
Replicate: binomial tree, 2,000 steps	Part 1	\$7.390	0.0009
Hedge: the PDE on a 400 × 400 grid	Part 2	\$7.390	0.0003
Average: Monte Carlo, 1,000,000 paths	Part 3	\$7.387	-0.0022
The Black-Scholes formula		\$7.390	0.0000

Part 1 copied the option on a tree; Part 2 hedged it and solved the Black-Scholes equation; Part 3 averages its payoff under risk-neutral odds. Every road lands on the same price within its own error.

Feynman-Kac

It isn't a coincidence. The **Feynman-Kac** theorem (Kac, 1949, building on Feynman's path integrals) says that the solution of a PDE of the Black-Scholes type with a payoff at expiry,

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0, \quad V(S, T) = \text{payoff}(S),$$

is an expectation over paths that grow at r with volatility σ :

$$V(S, t) = e^{-r(T-t)} \mathbb{E}^Q[\text{payoff}(S_T) \mid S_t = S].$$

Hedging and averaging are one statement written two ways; the tree is a step-by-step version of both. The notebook checks it at share prices of \$80, \$100 and \$120.

Which road, when?

Trees and grids step backwards in time, so at every point they can compare holding with exercising: the natural tools for American options. Our one-year \$105 American put is \$9.243 on the tree and \$9.241 on the grid, \$0.57 more than the European \$8.677. Simulation steps forward, so it shines for payoffs that depend on the whole path, or on many assets. And when there is a formula, use it.

The shared assumption

All three roads assume one volatility, the same for every strike and every day, and log returns shaped exactly like a bell curve. The next chapter tests that against real data.

Bip's rule. Replicate, hedge or average: one price.

Try this. An option pays on the *average* share price over the year. Which road would you take?

Sources

Companion notebook: `s3e24_three_roads.ipynb`. Every number in this chapter is computed by it.

1. American Mathematical Society, *On distributions of certain Wiener functionals* (M. Kac), *Transactions of the American Mathematical Society* 65(1), 1-13 (1949). <https://doi.org/10.1090/S0002-9947-1949-0027960-X>

The Smile: Where Black-Scholes Breaks

Real returns have fat tails, so the market prices each strike at its own volatility. FX quotes that smile as at-the-money, risk reversals and butterflies, by delta.

Last chapter's question

Monte Carlo: the payoff depends on the whole path, and simulation walks forward along it.

Fat tails in AUD/USD

Measure each daily move of the Australian dollar, 2 January 2018 to 18 September 2026, in standard deviations, and count the big ones against what a bell curve predicts for 2,176 days:

AUD/USD daily log changes. Source: Federal Reserve Board, H.10.

DAILY MOVES BEYOND	DAYS OBSERVED	A BELL CURVE PREDICTS
3 standard deviations	20	5.875
4 standard deviations	8	0.138
5 standard deviations	2	0.001

Big moves happen far more often than Black-Scholes allows (the excess kurtosis is 3.6; a bell curve has 0). The biggest, on 4 April 2025, was a fall of 5.0% at the Fed's noon rate: 7.9 standard deviations. The Reserve Bank of Australia described early April 2025 as a sharp rise in FX volatility as markets absorbed US tariff announcements, with the Australian dollar's largest daily decline (4.5 per cent, on the RBA's measure) outside the global financial crisis, and FX option volatility rising to pandemic levels.

One volatility per strike

If big moves are likelier than the bell curve says, options far from the money, which only pay after a big move, are worth more than one volatility would price them. The market quotes them at higher volatilities: a different volatility at every strike, the **smile**.

How FX options are quoted

FX desks quote by **delta** rather than strike, for each expiry: the **at-the-money** vol (here the delta-neutral straddle), the 25- and 10-delta **risk reversals** (call vol minus put vol; negative means AUD puts are dearer) and **butterflies** (how far the wings sit above at-the-money). For AUD/USD the premium is in US dollars and deltas are regular spot deltas out to one year (Reiswich and Wystup 2009, citing Clark). Our quotes are illustrative:

Illustrative AUD/USD volatility quotes, vol %.

EXPIRY	ATM	25-DELTA RR	25-DELTA BF	10-DELTA RR	10-DELTA BF
1M	8.40	-0.90	0.20	-1.70	0.65
3M	9.00	-1.20	0.25	-2.30	0.85
6M	9.30	-1.40	0.30	-2.70	1.00
1Y	9.60	-1.60	0.35	-3.10	1.15

From quotes to strikes

The simplified strangle turns quotes into vols, $\sigma_{25C} = \text{ATM} + \text{BF} + \text{RR}/2$ and $\sigma_{25P} = \text{ATM} + \text{BF} - \text{RR}/2$ (dealers' "market strangle" adds a refinement we leave out). Each delta is then a strike in disguise: solve $P_{AUD}N(d_1) = \Delta$ for d_1 and set $K = Fe^{-d_1\sigma\sqrt{T}+\sigma^2T/2}$; the delta-neutral at-the-money strike is $K = Fe^{\sigma^2T/2}$.

Three months: forward 0.70956. Strikes checked against QuantLib's BlackDeltaCalculator.

	VOL (%)	STRIKE	SPOT DELTA
10-delta put	11.00	0.6616	-0.100
25-delta put	9.85	0.6871	-0.250
at-the-money	9.00	0.7103	0.495
25-delta call	8.65	0.7314	0.250
10-delta call	8.70	0.7516	0.100

The first version of the notebook's ATM formula was wrong (it gave the at-the-money-forward strike); QuantLib's calculator disagreed by 0.00075, and the check caught it.

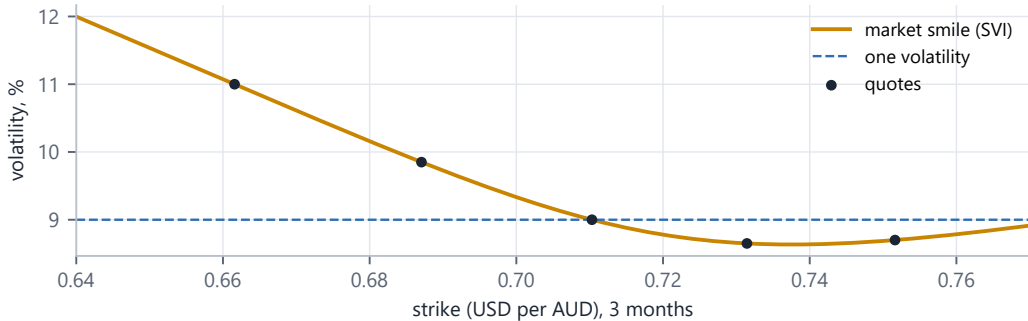


Figure 4.1. The 3-month smile through the five quotes, against one flat volatility.

Implied volatility is therefore not the market's forecast of volatility: one currency can't have five futures. It is a price of insurance, strike by strike.

Bip's rule. One volatility per strike: the smile is the market correcting the bell curve.

Try this. The 25-delta risk reversal moves from -1.2 to -2.0. Are AUD puts getting dearer or cheaper than calls?

Sources

Companion notebook: `s3e25_the_smile.ipynb`. Every number in this chapter is computed by it.

1. Board of Governors of the Federal Reserve System, *H.10 Foreign Exchange Rates, daily data download (Australian dollar, series R_{XI}\$US_N.B.AL)*. <https://www.federalreserve.gov/releases/h10/>
2. Reserve Bank of Australia, *Australia's External Position and the Evolution of the FX Markets (C. Kent, speech, 29 April 2025)*. <https://www.rba.gov.au/speeches/2025/sp-ag-2025-04-29.html>
3. Frankfurt School of Finance & Management (EconStor), *FX Volatility Smile Construction (D. Reiswich and U. Wystup), CPQF Working Paper No. 20 (2009)*. <https://www.econstor.eu/bitstream/10419/40186/1/613825101.pdf>

Building an Arbitrage-Free Surface: SVI

Fit each expiry with a five-parameter curve, then check that no butterfly and no calendar spread is free money.

Last chapter's question

Dearer: the 25-delta put's vol rises from 9.85% to 10.25% while the call's falls to 8.25%.

SVI

SVI ("stochastic volatility inspired", devised at Merrill Lynch in 1999 and made public by Jim Gatheral in 2004) writes the total variance $w = \sigma^2 T$ against log-moneyness $k = \ln(K/F)$:

$$w(k) = a + b \left(\rho (k - m) + \sqrt{(k - m)^2 + s^2} \right)$$

Five parameters with one job each: level a , wing steepness b , tilt ρ , shift m , rounding s . Far from the money it becomes straight lines, as option wings must. QuantLib's `SviInterpolatedSmileSection` fits it; with five quotes per expiry the fit is exact (largest error 7.8e-12 basis points of vol).

Fitted SVI parameters for each expiry.

EXPIRY	A	B	P	M	S
1M	0.00044	0.0056	-0.489	0.0058	0.0245
3M	0.00131	0.0136	-0.434	0.0140	0.0512
6M	0.00292	0.0195	-0.527	0.0178	0.0600
1Y	0.00684	0.0285	-0.619	0.0256	0.0689

Butterfly arbitrage

A butterfly (buy a call at $K - h$, sell two at K , buy one at $K + h$) pays a small tent and can never cost less than nothing. If a smile bends too sharply its prices imply a negative butterfly: free money. Gatheral and Jacquier (2014) turn this into a test on the total-variance smile:

$$g(k) = \left(1 - \frac{k w'}{2w}\right)^2 - \frac{w'^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \geq 0.$$

Their Example 3.1, from Axel Vogt, fails it: g dips to -0.033 for strikes from about 191% to 351% of the forward. Ours pass comfortably: the smallest g on a wide range of strikes is 0.0315.

Calendar arbitrage

At the same moneyness a longer option must be worth at least as much as a shorter one, so total variance can never fall with time. If it did, selling the short option and buying the long one would lock in money. Our expiries never cross; the closest pair is 0.0013 apart.

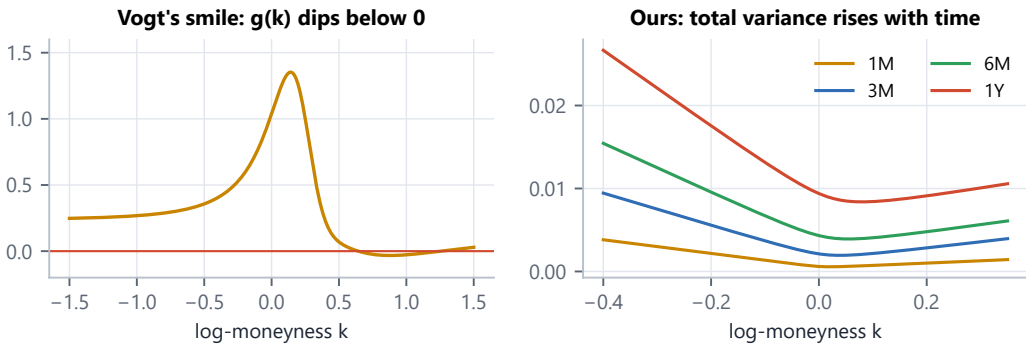


Figure 5.1. Left: Vogt's smile fails the butterfly test. Right: our four expiries' total variance, never crossing.

Bip's rule. A surface is only usable if it's arbitrage-free.

Try this. At some strike, the 3-month total variance is below the 1-month. Which arbitrage is that?

Sources

Companion notebook: `s3e26_svi_surface.ipynb`. Every number in this chapter is computed by it.

1. arXiv, *Arbitrage-free SVI volatility surfaces* (J. Gatheral and A. Jacquier), *Quantitative Finance* 14(1), 59-71 (2014); arXiv:1204.0646. <https://arxiv.org/abs/1204.0646>

What the Market Prices: Implied Densities

A butterfly's price is a probability, so the curvature of call prices across strikes is the market's risk-neutral density.

Last chapter's question

Calendar arbitrage: sell the short option, buy the long one.

Butterflies make a density

A butterfly around K pays a tent of area h^2 . Its price divided by h^2 is the discounted chance of landing near K , and as h shrinks it converges:

Butterflies around 0.71 on the 3-month AUD/USD smile.

WIDTH H	BUTTERFLY, USD PER AUD 1M	PRICE ÷ H ² ÷ DISCOUNT
0.020	5130.84	12.955
0.010	1329.42	13.426
0.005	335.48	13.553
0.001	13.46	13.594

In the limit (Breedon and Litzenberger, 1978):

$$\frac{\partial^2 C}{\partial K^2} = P_{USD} f(K).$$

The curvature of call prices across strikes *is* the discounted risk-neutral density.

AUD/USD in three months

On a fine strike grid the smile's density integrates to 1.000000, balances at 0.70956 (the forward is 0.70956), and is never negative beyond rounding: the surface passed Chapter 5's test.

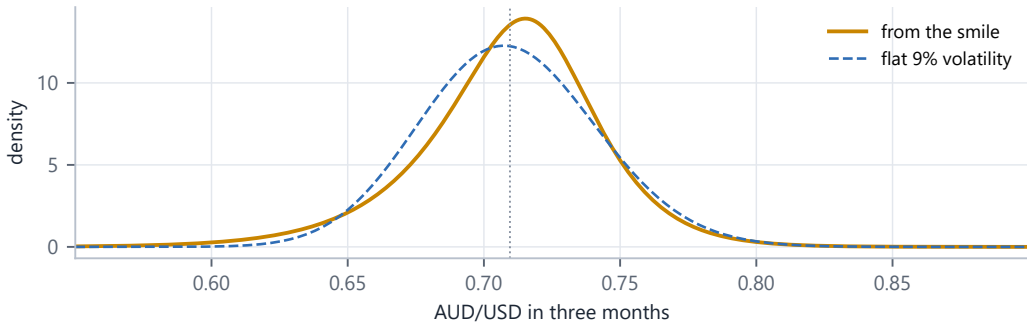


Figure 6.1. The implied density from the 3-month smile, against Black-Scholes' lognormal at one flat volatility.

Risk-neutral chances. The skew moves probability to the downside.

IN THREE MONTHS	WITH THE SMILE	FLAT VOLATILITY
AUD/USD below 0.65	5.1%	2.9%
AUD/USD above 0.75	9.7%	11.0%

A digital option

A digital call pays a fixed amount if AUD/USD ends above the strike. Its price is minus the *slope* of call prices, $-\partial C/\partial K = P_{USD} \Pr(S_T > K)$, so the smile's slope matters, not just its level. Above 0.72: 38.8% with the smile; 36.4% using the smile's vol at that strike but ignoring its slope; 36.7% with one flat volatility.

Not a forecast

These are risk-neutral odds (Chapter 2). They include what people will pay for insurance against a fall, so they give the price of each outcome, not its likelihood.

Bip's rule. The smile is a probability distribution in disguise.

Try this. Why must an implied density never be negative?

Sources

Companion notebook: `s3e27_implied_density.ipynb`. Every number in this chapter is computed by it.

1. University of Chicago Press, *Prices of state-contingent claims implicit in option prices* (D. T. Breeden and R. H. Litzenberger), *Journal of Business* 51(4), 621-651 (1978). <https://doi.org/10.1086/296025>

Local Volatility (Dupire)

Exactly one volatility function $\sigma(S, t)$ reproduces every vanilla on an arbitrage-free surface. It fits today perfectly, and predicts tomorrow's smile the wrong way round.

Last chapter's question

A negative density means a butterfly with a negative price: free money.

One volatility function

Keep Black-Scholes, but let volatility depend on where spot is and when: $dS = (r_{USD} - r_{AUD})S dt + \sigma_{loc}(S, t) S dW$. Dupire (1994) showed there is exactly one such function for an arbitrage-free surface. In total variance,

$$\sigma_{loc}^2 = \frac{\partial w / \partial T}{D(k)}$$

$$D(k) = 1 - \frac{k}{w} \frac{\partial w}{\partial k} + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{k^2}{w^2} \right) \left(\frac{\partial w}{\partial k} \right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial k^2}$$

The numerator needs no calendar arbitrage, the denominator is Chapter 5's $g(k)$: feed Dupire an arbitrage and the local variance comes out negative or infinite. It also differentiates the surface twice, so the notebook builds a smooth, dense one: the four SVI smiles, total variance linear in time between expiries, sampled monthly on a wide strike grid, with the curves re-expressed on ACT/365F because QuantLib's finite-difference engine reads curves by year fraction and Series 2's SOFR curve is ACT/360.

Local versus implied

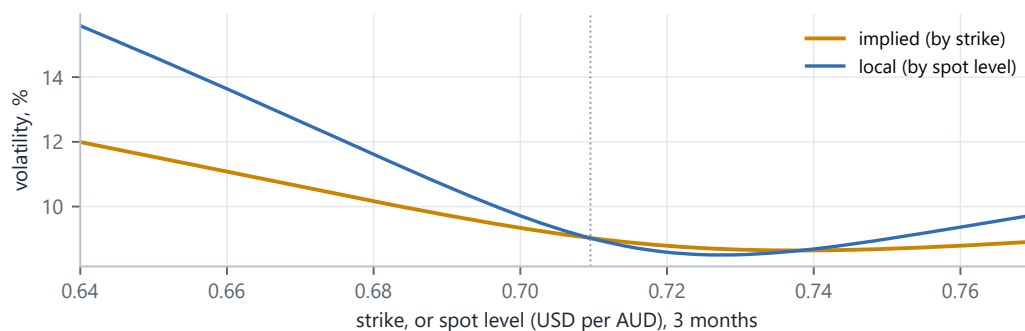


Figure 7.1. At three months: implied volatility by strike and local volatility by spot level.

Near the money they agree (9.02% local, 9.03% implied), but the local skew is 2.07 times the implied skew: implied vol is roughly an average of local vol along the way from spot to the strike.

An exact fit

USD per AUD 1 million. Local vol prices by finite differences (QuantLib, localVol=True).

	BLACK ON THE SURFACE	LOCAL VOL PDE	DIFFERENCE (BP OF VOL)
3M 25-delta put	5482.06	5482.48	0.037
3M at-the-money	12514.67	12514.97	0.021
6M 25-delta call	6487.85	6487.91	0.003
1Y 10-delta put	4407.66	4407.45	-0.017

The catch: smile dynamics

Freeze today's local volatility function and move spot: what does the model say tomorrow's at-the-money vol will be? (A subtlety: QuantLib's `LocalVolSurface` re-derives local vol from the implied surface whenever spot moves, which is really a smile frozen in strike. The notebook samples today's local vol into a `FixedLocalVolSurface` to test the model itself.)

3-month at-the-money-forward implied vol (%) after an instant spot move.

AUD/USD MOVES (%)	LOCAL VOL MODEL	SMILE FROZEN IN STRIKE	SMILE MOVES WITH SPOT
-3	11.08	9.80	9.03
0	9.03	9.03	9.03
3	8.69	8.66	9.03

After a 3% fall, local vol lifts the ATM vol by 205bp, far more than a smile frozen in strike (77bp), while FX desks often assume the smile simply moves with spot. Hagan, Kumar, Lesniewski and Woodward (2002) showed that local vol's smile moves the wrong way with spot, so its deltas, and its hedges, can be worse than plain Black-Scholes, even while every price matches.

Bip's rule. Local vol fits today's surface exactly, and gets tomorrow's wrong.

Try this. If local vol fits every price, why might its delta still be wrong?

Sources

Companion notebook: `s3e28_local_vol.ipynb`. Every number in this chapter is computed by it.

1. Risk, *Pricing with a smile* (B. Dupire), *Risk* 7(1), 18-20 (January 1994). <https://www.risk.net/derivatives/equity-derivatives/1500211/pricing-with-a-smile>
2. Wilmott, *Managing smile risk* (P. S. Hagan, D. Kumar, A. S. Lesniewski and D. E. Woodward), *Wilmott Magazine*, September 2002, 84-108. https://www.researchgate.net/publication/235622441_Managing_Smile_Risk

Stochastic Volatility: The Heston Model

Let volatility move by itself, pulled towards a long-run level and correlated with the currency. Correlation makes the skew, vol-of-vol the curvature.

Last chapter's question

Delta depends on how the smile moves when spot moves, and local vol gets that movement the wrong way round.

The model

Heston (1993):

$$dS = (r_{USD} - r_{AUD}) S dt + \sqrt{v} S dW_1$$

$$dv = \kappa(\theta - v) dt + \xi \sqrt{v} dW_2, \quad dW_1 dW_2 = \rho dt$$

Five parameters: today's variance v_0 , the long-run level θ it is pulled to at speed κ , the vol-of-vol ξ , and the correlation ρ . European options have a closed-form price, so calibration is fast.

Calibrated to AUD/USD

Fitted to all 20 quotes at once (QuantLib's `HestonModelHelper`, Levenberg-Marquardt, on the ACT/365F curves): volatility today 8.58%, long-run 11.54%, a half-life of 3.1 months, vol-of-vol 0.33, correlation -0.37. The Feller condition fails ($2\kappa\theta - \xi^2 = -0.0370$), which is common and harmless for the closed form.

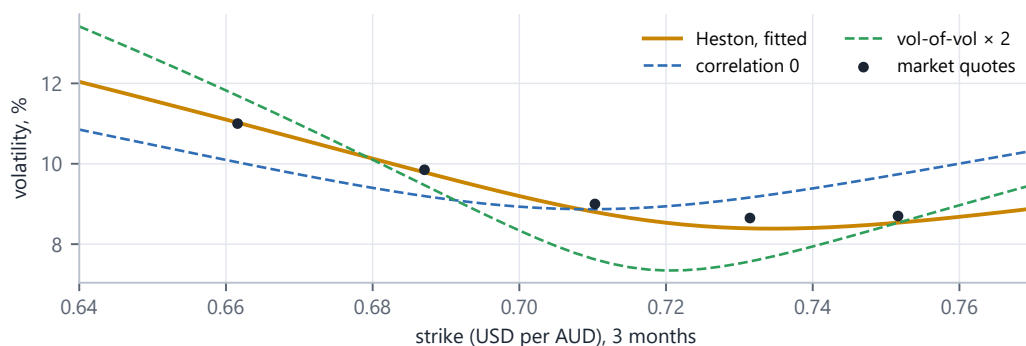


Figure 8.1. The 3-month smile from the fitted model, with the correlation set to zero, and with the vol-of-vol doubled.

Set the correlation to zero and the tilt disappears; double the vol-of-vol and the smile bends more at both ends.

The fit

Model minus market, basis points of volatility.

EXPIRY	10-DELTA PUT	25-DELTA PUT	ATM	25-DELTA CALL	10-DELTA CALL
1M	11.2	19.0	18.2	6.4	-3.1
3M	0.9	-6.1	-18.2	-23.8	-15.9
6M	-3.3	-6.8	-9.5	-11.7	-3.5
1Y	-15.8	-0.1	19.7	20.9	18.5

Five parameters can't match 20 quotes: errors reach 24bp, and they swing between expiries because one parameter set must serve all four. The closed form itself checks out: QuantLib's Monte Carlo Heston engine gives USD 11927 ± 55 per AUD 1 million for the 3-month at-the-money option, against 11966 from the formula, well within its error bar. (A trap on the way: `HestonModel.calibrate` doesn't update the process object you passed in, so a simulation must be built from the calibrated parameters.)

The trade-off

Local vol fits exactly with the wrong dynamics; Heston has sensible dynamics and an approximate fit. Many desks combine them: **stochastic local volatility**.

Bip's rule. Stochastic vol explains the smile; it doesn't fit it exactly.

Try this. Fit Heston with the correlation fixed at zero. What happens to the skew?

Sources

Companion notebook: `s3e29_heston.ipynb` . Every number in this chapter is computed by it.

1. Oxford University Press, *A closed-form solution for options with stochastic volatility with applications to bond and currency options* (S. L. Heston), *Review of Financial Studies* 6(2), 327-343 (1993). <https://doi.org/10.1093/rfs/6.2.327>

SABR: The Rates Desk's Smile

A closed-form smile from four parameters. Alpha sets the level, rho the skew, nu the curvature, and beta the backbone.

Last chapter's question

It disappears: with $\rho = 0$ the smile turns (nearly) symmetric.

The caplet smile

Part 2's caplet on 3-month BBSW, set one year from today on the Series 1 curve (forward 4.2488%), now with a volatility for every strike, in normal (basis-point) volatility, Bachelier's convention. The quotes are illustrative: generated from a SABR smile and rounded to 0.1bp, so a near-perfect fit is built in. Real smiles fit a little less exactly.

SABR

Hagan, Kumar, Lesniewski and Woodward (2002), in the same paper that criticised local vol:

$$dF = \alpha F^\beta dW_1, \quad d\alpha = \nu \alpha dW_2, \quad dW_1 dW_2 = \rho dt$$

α sets the level, ρ the skew, ν the curvature, and β the **backbone**: how volatility moves when rates move. Hagan's closed-form approximation gives the implied vol at every strike. Desks usually fix β and fit the other three; QuantLib's Python build has no SABR fitter, so the notebook uses `ql.Optimizer` with Simplex.

$\beta = 0.5$; $\alpha = 0.0356$, $\rho = -0.25$, $\nu = 0.40$.

STRIKE VS FORWARD (BP)	QUOTE (BP)	SABR (BP)	ERROR (BP)
-100	77.2	77.19	-0.011
-50	75.1	75.10	0.002
-25	74.4	74.43	0.030
0	74.1	74.09	-0.014
25	74.1	74.12	0.020
50	74.6	74.56	-0.044
100	76.6	76.62	0.017

Hagan's formula typed out by hand matches QuantLib's `sabrVolatility` to 8.3e-17.

The backbone

Move the forward with the parameters fixed and watch the at-the-money normal vol: 69.6bp if rates fall 50bp, 74.1 today, 78.3 if they rise 50bp. With $\beta = 0$ it barely moves; with $\beta = 1$ it rises in step with the rate.

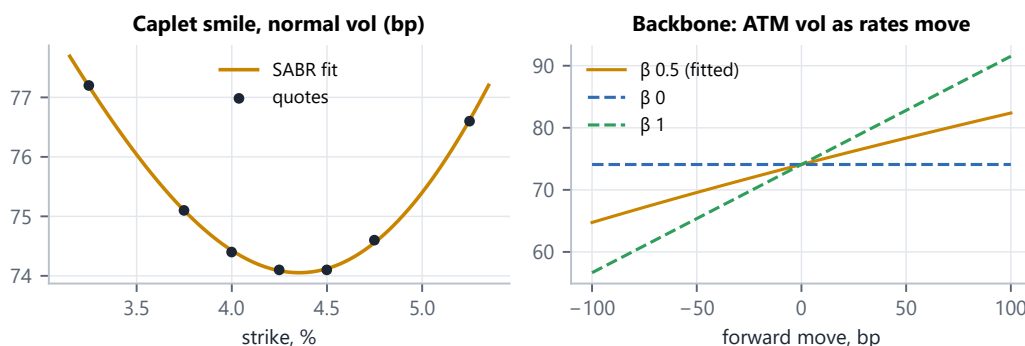


Figure 9.1. Left: the caplet smile and the SABR fit. Right: the backbone for three values of β .

Rates desks like SABR because it is fast, its knobs are readable, its dynamics are chosen through β , and it scales to one smile per expiry and underlying tenor. When rates can go negative, they shift the rate, or use $\beta = 0$.

Bip's rule. SABR: level, skew, curvature, and a backbone.

Try this. With $\beta = 0$ and $\nu = 0$, what model does SABR become?

Sources

Companion notebook: `s3e30_sabr.ipynb`. Every number in this chapter is computed by it.

1. Wilmott, *Managing smile risk* (P. S. Hagan, D. Kumar, A. S. Lesniewski and D. E. Woodward), *Wilmott Magazine*, September 2002, 84-108. https://www.researchgate.net/publication/235622441_Managing_Smile_Risk

An AUD/USD Options Book

Price the book on the surface, read its risks, hedge the big ones, and explain a week's P&L. Greeks tell you what to hedge; revaluation tells you what happened.

Last chapter's question

Bachelier's normal model (Part 2, Chapter 11): with $\beta = 0$ the forward moves by αdW , and with $\nu = 0$, α never changes.

The book

Priced with Black's formula on the forward at the smile's vol for each strike; each checked against QuantLib's `blackFormula`. Book value USD 50,724.

	STRIKE	VOL (%)	VALUE (USD)
Long 3M 25-delta AUD put	0.6871	9.85	+54,774
Short 1M ATM straddle: call	0.7101	8.40	-34,034
Short 1M ATM straddle: put	0.7101	8.40	-35,094
Long 6M 0.7200 AUD call	0.7200	9.11	+65,078

Insurance against a fall (the 3-month put), a bet on calm (the short 1-month straddle) and a longer-dated call.

The risks

Bump and revalue on the full surface, as a desk's risk system does:

The book's risks.

Delta	-141,727 AUD
Gamma (delta change per 1% fall)	896,988 AUD
Vega, 1M	-8,244 USD per vol pt
Vega, 3M	10,428 USD per vol pt
Vega, 6M	9,783 USD per vol pt
Skew (RR 1 pt more negative)	2,614 USD
Theta, one week	2,763 USD

Short gamma at one month (a fall makes the book longer AUD, fast), long vega at three and six months, long skew, and time working in its favour.

Hedges

Buy about AUD 141,700 spot for delta. Flatten the 3-month vega with a tool that has vega and almost no delta: sell about AUD 3.6 million of 3-month at-the-money straddles. Keep the short 1-month gamma and the long skew: that's the book's view.

The week

Seven days pass; AUD/USD falls 1.5%, at-the-money vols rise half a point, and the 25-delta risk reversals move 0.3 of a point more negative. With the smiles rebuilt from the moved quotes around the new forward (the smile moves with spot: "sticky delta"), the option book makes USD 3,303.

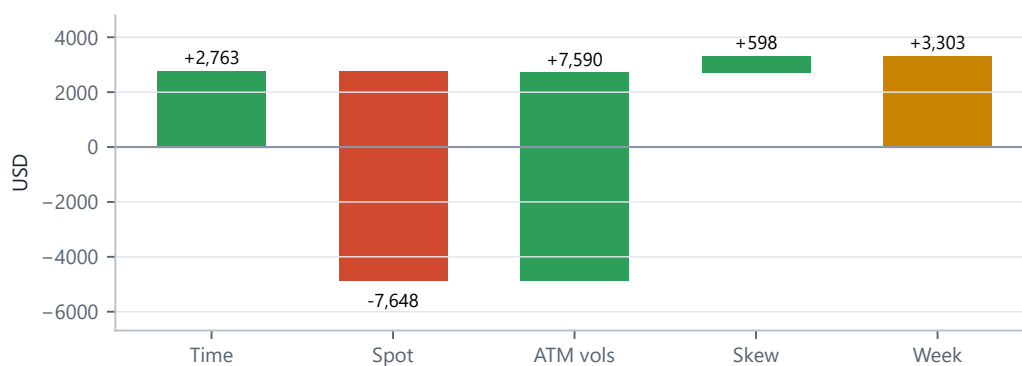


Figure 10.1. The week explained by full revaluation, one change at a time.

Step by step: time USD 2,763, spot –USD 7,648, ATM vols USD 7,590, skew USD 598: exactly USD 3,303.

Greeks versus reality

From Monday's Greeks alone (delta, gamma and vanna for spot, vega for the vol moves, the skew sensitivity, a week's theta) the estimate is USD 4,594, off by USD 1,291. The main reason is the short 1-month straddle: the same spot fall would have cost USD 5,254 on Monday but cost USD 7,648 a week later, because gamma grows as expiry nears (Part 2, Chapter 5). Greeks are a snapshot.

The series in one picture

Part 1 answered why our call is worth \$7.39 by copying it on a tree. Part 2 reached the same price by hedging, built the risk dials and took the formula into AUD/USD and BBSW. Part 3 found the third road, watched Black-Scholes break on real data, and rebuilt it with the models desks use.

Bip's rule. Greeks tell you what to hedge; revaluation tells you what happened.

Try this. The final checkpoint in the appendix.

Companion notebook: `s3e31_capstone.ipynb` . Every number in this chapter is computed by it.

Final checkpoint and glossary

Final checkpoint

Try these before looking at the answers.

1. A path's likelihood weight is 1.3. Is that path likelier under the real-world or the risk-neutral odds?
2. The 25-delta risk reversal is -1.2 and the butterfly 0.25, with at-the-money at 9.0. What are the 25-delta call and put vols?
3. Why can't a surface whose total variance falls with time be used for local vol?
4. Your book is short gamma and long vega. What kind of week hurts most?

Answers.

1. Under the risk-neutral odds. The weight is the ratio of risk-neutral to real-world probability, so a weight above 1 means the risk-neutral odds make the path more likely (typically a low-return path).
2. Call: $9.0 + 0.25 - 0.6 = 8.65\%$. Put: $9.0 + 0.25 + 0.6 = 9.85\%$.
3. Dupire's numerator $\partial w / \partial T$ would be negative, giving negative local variance: the surface has calendar arbitrage.
4. A big move in spot while implied volatilities fall: the short gamma loses on the move, and the long vega loses on the vol drop.

Glossary

Butterfly (FX quote). How far the wing vols sit above at-the-money, on average.

Butterfly arbitrage. A smile whose prices imply a butterfly spread with a negative price; equivalently a negative implied density, or $g(k) < 0$.

Calendar arbitrage. Total variance falling with time at some moneyness.

Delta-neutral straddle (ATM). The strike at which the call and put deltas cancel; the at-the-money convention used here for AUD/USD.

Digital option. Pays a fixed amount if the underlying ends above (or below) the strike.

Dupire's formula. Local volatility from an arbitrage-free implied surface.

Fat tails. More large moves than a bell curve allows; positive excess kurtosis.

Feynman-Kac. The solution of a Black-Scholes-type PDE is a discounted expectation.

Girsanov's theorem. Changing the probability measure by path weights changes the drift of a Brownian motion but not its volatility.

Heston model. Stochastic variance with mean reversion, vol-of-vol and correlation to the asset.

Implied density. The risk-neutral distribution read from option prices: the curvature of call prices across strikes (Breedon and Litzenberger).

Local volatility. Volatility as a function of spot and time, fitted to reproduce every vanilla.

Market price of risk. Extra expected return per unit of volatility, $(\mu - r)/\sigma$.

Martingale. A process whose expected future value is its value today.

Risk reversal. Call vol minus put vol at the same delta.

SABR. A stochastic volatility model for a forward with a closed-form smile; β sets its backbone.

Smile, skew. Implied volatility varying with strike; a skew is a lopsided smile.

Sticky delta, sticky strike. Assumptions about how the smile moves when spot moves: with spot, or fixed in strike.

SVI. A five-parameter smile for total variance.

Total variance. Implied variance times time, $w = \sigma^2 T$.

Vega bucket. Sensitivity to the volatility of one expiry.

Further reading

These books cover the same ground from different angles; none of this book's examples come from them.

- Jim Gatheral, *The Volatility Surface: A Practitioner's Guide*: the smile, SVI, local and stochastic volatility.
- Iain J. Clark, *Foreign Exchange Option Pricing: A Practitioner's Guide*: FX smile conventions and models in depth.

- Steven E. Shreve, *Stochastic Calculus for Finance II*: Girsanov, Feynman-Kac and the rigorous foundations.
- Fabrice Douglas Rouah, *The Heston Model and Its Extensions in Matlab and C#*: Heston in detail.
- Lorenzo Bergomi, *Stochastic Volatility Modeling*: smile dynamics, and why they matter for hedging.